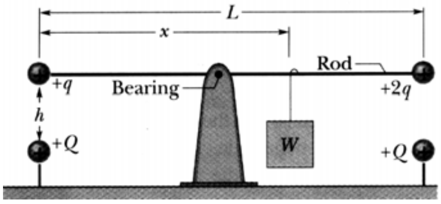


1. Die figuur toon 'n lang, nie-geleidende massalose staaf met lengte L wat roteer om sy middel en gebalanseer word deur 'n gewig W op 'n afstand x vanaf die linkerkant. Aan die linker en regterkante is klein geleidende sfere met ladings van q en $2q$ vas. 'n Afstand h onder elkeen is vaste sfere met ladings Q. Bereken die afstand x wanneer die staaf horisontaal en gebalanseer is.

The figure shows a long, nonconducting, massless rod of length L, pivoted at its center and balanced with a weight W at a distance x from the left. At the left and right ends of the rod are attached small conducting spheres with charges q and 2q. A distance h directly beneath each is a fixed sphere with charge Q. Find the distance x when the rod is horizontal and balanced. . (6)

$$\sum \tau = 0 \quad \therefore -\frac{1}{4\pi\varepsilon_0}\frac{qQ}{h^2}\left(\frac{L}{2}\right)+\frac{1}{4\pi\varepsilon_0}\frac{2qQ}{h^2}\left(\frac{L}{2}\right)-W\left(x-\frac{L}{2}\right)=0$$

$$\frac{L}{2}\left[\frac{1}{4\pi\varepsilon_0}\frac{qQ}{h^2}+W\right]=Wx$$

$$\therefore x=\frac{L}{2}\left[\frac{1}{4\pi\varepsilon_0}\frac{qQ}{h^2W}+1\right]$$

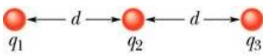
Vir watter waarde van h sal daar geen netto vertikale krag wees as die staaf horisontaal gebalanseer is?

For what value of h will there be no vertical force when the rod is horizontally balanced? (3)

$$\sum F = 0 \ : \quad \frac{1}{4\pi\varepsilon_0}\frac{qQ}{h^2}+\frac{1}{4\pi\varepsilon_0}\frac{2qQ}{h^2}-W=0$$

$$\frac{3}{4\pi\varepsilon_0}\frac{qQ}{1}=Wh^2$$

$$\therefore h=\sqrt{\frac{3}{4\pi\varepsilon_0}\frac{qQ}{W}}$$



2. Die drie ladings in die figuur lê in 'n reguit lyn en is 'n afstand d uitmekaar. Ladings q_1 en q_2 is vas terwyl lading q_3 vry is om te beweeg maar is in ekwilibrium (geen netto krag werk daarop). Bepaal q_1 in terme van q_2.

In the figure, three charged particles lie on a straight line and are separated by distances d. Charges q_1 and q_2 are held fixed. Charge q_3 is free to move but happens to be in equilibrium (no net electrostatic force on it). Find q_1 in terms of q_2. (4)

$$k\frac{q_1q_3}{(2d)^2}+k\frac{q_2q_3}{d^2}=0$$

$$\frac{q_1}{4}+q_2=0$$

$$q_1=-4q_2$$

3. 'n Positiewe punt lading met 'n grootte van 4.00×10^{-8} C word in 'n reghoekige koördinaat sisteem by 'n punt $x = +0.10$ m, $y = 0$ geplaas. 'n Identiese punt lading word by 'n punt $x = -0.10$ m, $y = 0$ geplaas. Bepaal die grootte en rigting van die elektriese veld by die punt $x = 0.10$ m, $y = 0.15$ m.
In a rectangular coordinate system a positive point charge with a magnitude of 4.00×10^{-8} C is placed at a point $x = +0.10$ m, $y = 0$. An identical point charge is placed at $x = -0.1$ m, $y = 0$. Calculate the magnitude and direction of the electric field at the point $x = 0.10$ m, $y = 0.15$ m. (10)

$$E_1 = \frac{(8.99\times10^9)(4\times10^{-8})}{(0.15)^2} = 1.6\times10^4 \quad \text{N/C}$$

$$\theta_1 = 90°$$

$$r_2 = \sqrt{0.15^2 + 0.2^2} = 0.25 \text{ m}$$

$$E_2 = \frac{(8.99\times10^9)(4\times10^{-8})}{(0.25)^2} = 5.76\times10^3 \quad \text{N/C}$$

$$\theta_2 = \sin^{-1}\left(\frac{0.15}{0.25}\right) = 36.87°$$

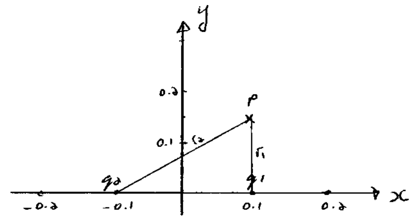
$$E_{2x} = 5.76\times10^3 \cos 36.87° = 4.61\times10^3 \quad \text{N/C}$$

$$E_{2y} = 5.76\times10^3 \sin 36.87° = 3.46\times10^3 \quad \text{N/C}$$

$$\overline{E} = (E_{2x})\hat{x} + (E_{1y} + E_{2y})\hat{y} = (4.61\times10^3\,\hat{x} + 1.95\times10^4\,\hat{y}) \text{ N/C}$$

$$E = \sqrt{(4.61\times10^3)^2 + (1.95\times10^4)^2} = 2\times10^4 \quad \text{N/C}$$

$$\text{rigting} = \tan^{-1}\left(\frac{1.95}{0.46}\right) = 76.67°$$



4. 'n Voorwerp met 'n massa van 10.0 g en 'n lading van 8.00×10^{-5} C word in 'n elektriese veld van $E_x=3\times10^3$ N/C, $E_y=-600$ N/C en $E_Z=0$ geplaas. Wat is die grootte en rigting van die krag op die voorwerp? *An object having a mass of 10.0 g and a charge of 8.00×10^{-5} C is placed in an electric field with $E_X=3\times10^3$ N/C, $E_Y=-600$ N/C and $E_Z=0$. (a) What are the magnitude and direction of the force on the object?* (6)

$$\vec{F} = q\vec{E} = (8\times10^{-5})(3.0\times10^3\,\hat{i} - 600\,\hat{j}) = (0.24\,\hat{i} - 0.048\,\hat{j})$$

$$F = \sqrt{F_x^2 + F_y^2} = \sqrt{0.24^2 + 0.048^2} = 0.245 \quad \text{N}$$

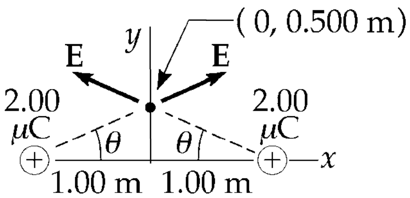
$$\theta = \tan^{-1}\left(\frac{F_y}{F_x}\right) = \tan^{-1}\left(\frac{-0.048}{0.24}\right) = -11.3°$$

As die voorwerp uit rus vanaf die oorsprong losgelaat word, wat sal sy koördinate na 3.00 s wees.?
If the object is released from rest at the origin, what will be its coordinates after 3.00 s? (6)

$$x = \tfrac{1}{2} a_x t^2 = \tfrac{1}{2}\frac{F_x}{m} t^2$$
$$= \tfrac{1}{2}\frac{0.24}{1\times 10^{-2}}(3)^2$$
$$= 108 \text{ m}$$

$$y = \tfrac{1}{2} a_y t^2 = \tfrac{1}{2}\frac{F_y}{m} t^2$$
$$= \tfrac{1}{2}\frac{-0.048}{1\times 10^{-2}}(3)^2$$
$$= -21.6 \text{ m}$$

5. Twee 2.00 μC punt ladings is op die x-as, een by x = 1.00 m en die ander by x = -1.00 m. Bepaal die elektriese veld op die y-as by y = 0.50 m.
Two 2.00 μC point charges are located on the x-axis. One is at x = 1.00 m and the other is at x = -1.00 m. Determine the electric field on the y-axis at y = 0.50 m. (6)



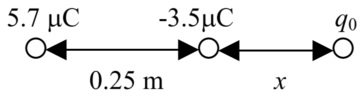
$$E = \frac{k_e q}{r^2}$$
$$= \frac{(8.99\times 10^{9})(2.00\times 10^{-6})}{(1.12)^2}$$
$$= 14400 \text{ N/C}$$

$$E_x = 0$$
$$E_y = 2(14{,}400)\sin 26.6°$$
$$= 1.29\times 10^4 \text{ N/C}$$
$$\vec{E} = 1.29\times 10^4\,\hat{j} \text{ N/C}$$

Bepaal die elektriese krag op 'n –3.00 μC lading wat op die y-as by y = 0.50 m geplaas word.
Calculate the electric force on a –3.00 μC charge placed on the y-axis at y = 0.50 m. (2)

$$\vec{F} = \vec{E}q$$
$$= (1.29\times 10^4\,\hat{j})(-3.00\times 10^{-6})$$
$$= -3.86\times 10^{-2}\,\hat{j} \text{ N}$$

6. ’n +5.7 μC en ’n -3.5 μC lading is 25 cm van mekaar. Waar kan ’n derde lading geplaas word sodat daar geen eksterne krag daarop is nie?
A +5.7 μC and a -3.5 μC charge are placed 25 cm apart. Where can a third charge be placed so that it experiences no net force? (7)



$$\vec{F}_1 = -\vec{F}_2$$

$$\frac{k(5.7\times 10^{-6})}{(x+0.25)^2} = \frac{k(3.5\times 10^{-6})}{x^2}$$

$$5.7x^2 = 3.5(x+0.25)^2$$

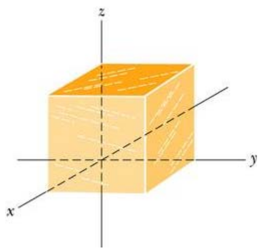
$$5.7x^2 - 3.5(x^2 + 0.5x + 0.0625) = 0$$

$$2.2x^2 - 1.75x - 0.218 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{1.75 \pm \sqrt{1.75^2 - 4(2.2)(-0.218)}}{2(2.2)}$$

$$= 0.905 \text{ m (-11 cm nie geldig nie)}$$



7. Bepaal die netto vloed deur die kubus met sylengte 1.40 m as die elektriese veld gegee word deur $\vec{E} = 3.00y\,\hat{\mathrm{j}}$.
Find the net flux through the cube with edge lengths 1.40 m if the electric field is given by $\vec{E} = 3.00\,y\,\hat{\mathrm{j}}$. (3)

$$\Phi = (3.00y\,\hat{\mathrm{j}})\cdot(\text{-}A\,\hat{\mathrm{j}})\Big|_{y=0} + (3.00y\,\hat{\mathrm{j}})\cdot(A\,\hat{\mathrm{j}})\Big|_{y=1.4}$$

$$= (3.00)(1.4)(1.4)^2$$

$$= 8.23 \text{ N m}^2/\text{C}$$

Bepaal die netto vloed deur die kubus as die elektriese veld gegee word deur $\vec{E} = -4.00\,\hat{\mathrm{i}} + (6.00 + 3.00y)\hat{\mathrm{j}}$.
Find the net flux through the cube if the electric field is given by $\vec{E} = -4.00\,\hat{\mathrm{i}} + (6.00 + 3.00\,y)\,\hat{\mathrm{j}}$. (3)

$$\vec{E} = (-4.00\,\hat{\mathrm{i}} + 6.00\,\hat{\mathrm{j}}) + 3.00\,y\,\hat{\mathrm{j}}$$

$$\vec{E} = (-4.00\,\hat{\mathrm{i}} + 6.00\,\hat{\mathrm{j}}) \equiv \text{constant field} \Rightarrow \text{no flux through cube}$$

$$\text{flux for } \vec{E} = 3.00y\,\hat{\mathrm{j}} \Rightarrow \Phi = 8.23\text{ Nm}^2/\text{C}$$

Hoeveel lading word in elk van bostaande gevalle deur die kubus omsluit?
In each of the above cases, how much charge is enclosed by the cube? (2)

$$q = \varepsilon_0 \Phi$$

$$= 8.85\times 10^{12}(8.23)$$

$$= 7.29\times 10^{-11} \text{ C}$$

8. ‘n Punt lading van 1.8 μC is by die middelpunt van ‘n kubiese Gaussian oppervlak met sylengte 55 cm. Wat is die netto elektrise vloed deur die oppervlak?
A point charge of 1.8 μC is at the center of a cubic Gaussian surface with edge length of 55 cm. What is the net electric flux through the surface? (3)

$$\Phi = \frac{q}{\varepsilon_0} = \frac{1.8\times10^{-6}}{8.85\times10^{-12}}$$
$$= 2.0\times10^{5} \quad \mathrm{N\,m^2/C}$$

9. ‘n Oneindige lyn van lading produseer ‘n veld van 4.5×10^4 N/C op ‘n afstand van 2.0 m. Bepaal die liniêre ladings digtheid λ.
An infinite line of charge produces a field of 4.5×10^4 N/C at a distance of 2.0 m. Calculate the linear charge density λ. (3)

$$\lambda = 2\pi\,\varepsilon_0 E r$$
$$= 2\pi\,(8.85\times10^{-12})(4.5\times10^{4})(2.0)$$
$$= 5\times10^{-6} \quad \mathrm{C/m}$$

10. Veronderstel ’n positiewe lading is uniform versprei oor ’n baie lang silindriese volume met radius *R* en lading per eenheids volume van ρ. Lei ’n uitdrukking af vir die elektriese veld binne die volume op ’n afstand *r* vanaf die as van die silinder in terme van die ladings digtheid ρ.
Suppose positive charge is uniformly distributed throughout a very long cylindrical volume with radius R with charge per unit volume ρ. Derive the expression for the electric field inside the volume at a distance r from the axis of the cylinder in terms of the charge density ρ. (4)

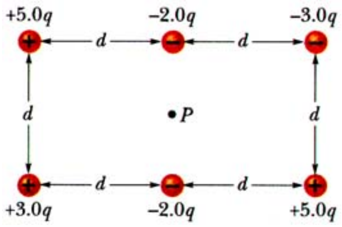
$$\Phi = \oint \vec{E}\cdot d\vec{A} = E 2\pi\, r l = q_{\mathrm{enc}}/\varepsilon_0$$
$$\text{nou}: q_{\mathrm{enc}} = \rho\pi r^2 l$$
$$\therefore E2\pi r l = \rho\pi r^2 l/\varepsilon_0$$
$$E = \rho r/2\varepsilon_0$$

Wat is die elektriese veld by ’n punt buite die volume in terme van die lading per eenheids lengte λ in die silinder?
What is the electric field at a point outside the volume in terms of the charge per unit length λ in the cylinder? (3)

$$q'_{\mathrm{enc}} = \lambda l$$
$$\therefore E2\pi r l = \lambda l/\varepsilon_0$$
$$\therefore E = \lambda/2\pi r\varepsilon_0$$

Verkry die verband tussen ρ en λ deur van die antwoorde hierbo gebruik te maak (by $r = R$).
Derive the relationship between ρ and λ by using the answers above (at r= R). (3)

$$\frac{\rho R}{2\varepsilon_0} = \frac{\lambda}{2\pi R\varepsilon_0}$$
$$\therefore \rho = \frac{\lambda}{\pi R^2}$$



11. Punt P is by die middel van die reghoek. Wat is die netto elektriese potensiaal by P a.g.v. die ses ladings?.

Point P is at the centre of the rectangle. What is the net electric potential at P due to the six charges? (8)

$$V = \sum_{i=1}^{6} V_{Pi} = \sum_{i=1}^{6} \frac{q_i}{4\pi\varepsilon_0 r_i}$$

$$= \frac{1}{4\pi\varepsilon_0}\left[\frac{5q}{\sqrt{d^2+(d/2)^2}} + \frac{-2q}{d/2} + \frac{-3q}{\sqrt{d^2+(d/2)^2}} + \frac{+3q}{\sqrt{d^2+(d/2)^2}} + \frac{-2q}{d/2} + \frac{+5q}{\sqrt{d^2+(d/2)^2}}\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\left[\frac{5q}{\frac{\sqrt{5}}{2}d} - \frac{2q}{\frac{d}{2}} - \frac{3q}{\frac{\sqrt{5}}{2}d} + \frac{3q}{\frac{\sqrt{5}}{2}d} - \frac{2q}{\frac{d}{2}} + \frac{5q}{\frac{\sqrt{5}}{2}d}\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\left[\frac{10q}{\sqrt{5}d} - \frac{4q}{d} - \frac{6q}{\sqrt{5}d} + \frac{6q}{\sqrt{5}d} - \frac{4q}{d} + \frac{10q}{\sqrt{5}d}\right] = \frac{0.94q}{4\pi\varepsilon_0 d}$$

12. Veronderstel ‘n elektron in die buis van ‘n TV stel word vanuit rus versnel deur ‘n potensiaal verskil van 5000 V. Wat is die verandering in die potensiële energie van die elektron?
Suppose an electron in the tube of a TV set is accelerated from rest through a potential difference of 5000 V. What is the change in the potential energy of the electron? (2)

$$\begin{aligned}\Delta U &= qV \\ &= (-1.6\times10^{-19})(5000) \\ &= -8.0\times10^{-16} \quad \text{J}\end{aligned}$$

Wat is die spoed van die elektron (m = 9.1×10^{-31}) kg as gevolg van hierdie versnelling?
What is the speed of the electron (m = 9.1×10^{-31}) kg *as a result of this acceleration?* (3)

$$\begin{aligned}\tfrac{1}{2}mv^2 &= qV \\ v &= \sqrt{\frac{2qV}{m}} \\ &= \sqrt{\frac{2(1.6\times10^{-19})(5000)}{9.1\times10^{-31}}} \\ &= 4.19\times10^{7}\ \text{m/s}\end{aligned}$$

Herhaal bostaande twee vrae vir ‘n proton (m = 1.67×10^{-27} kg) wat deur V = -5000 V versnel word.
Repeat the above two questions for a proton (m = 1.67×10^{-27} kg) that accelerates through V = -5000 V. (5)

$$\begin{aligned}\Delta U &= qV \\ &= (1.6\times10^{-19})(-5000) \\ &= -8.0\times10^{-16}\ \text{J} \\ v &= \sqrt{\frac{2qV}{m}} \\ &= \sqrt{\frac{2(1.6\times10^{-19})(+5000)}{1.67\times10^{-27}}} \\ &= 9.79\times10^{5}\quad \text{m/s}\end{aligned}$$

13. Die elektriese potensiaal binne ‘n gelaaide sferiese geleier met radius R word gegee deur $V = k_eQ/R$ en buite die geleier deur $V = k_eQ/r$. Gebruik $E_r = -dV/dr$, en lei ‘n uitdrukking af vir die elektriese veld binne hierdie ladings verspreiding.
The electric potential inside a charged spherical conductor of radius R is given $V = k_eQ/R$ and outside the conductor by $V = k_eQ/r$. Using $E_r = -dV/dr$, derive an expression for the electric field inside this charge distribution. (2)

$$\text{For } r < R \qquad V = \frac{k_e Q}{R}$$

$$E_r = -\frac{dV}{dr} = 0$$

Lei ‘n uitdrukking af vir die elektriese veld buite hierdie ladings verspreiding.
Derive an expression for the electric field outside this charge distribution. (3)

$$\text{For } \quad r \geq R \qquad V = \frac{k_e Q}{r}$$

$$\begin{aligned} E_r &= -\frac{dV}{dr} = -\left(-\frac{k_e Q}{r^2}\right) \\ &= \frac{k_e Q}{r^2} \end{aligned}$$

14. Toon aan dat die hoeveelheid werk wat benodig word om vier identiese puntladings met grootte Q by die hoeke van ‘n vierkant met sylengte s bymekaar te bring gegee word deur 5.14 k_e Q^2/s.
Show that the amount of work required to assemble four identical point charges of magnitude Q at the corners of a square of side s is 5.14 k_e Q^2/s. (6)

$$\begin{aligned} U &= U_1 + U_2 + U_3 + U_4 \\ &= 0 + U_{12} + (U_{13} + U_{23}) + (U_{14} + U_{24} + U_{34}) \\ &= 0 + \frac{k_e Q^2}{s} + \frac{k_e Q^2}{s}\left(\frac{1}{\sqrt{2}} + 1\right) + \frac{k_e Q^2}{s}\left(1 + \frac{1}{\sqrt{2}} + 1\right) \\ &= \frac{k_e Q^2}{s}\left(4 + \frac{2}{\sqrt{2}}\right) \\ &= 5.41\frac{k_e Q^2}{s} \end{aligned}$$